

JavaScript Detailed Notes & Important Questions

■ Important Question: What is JavaScript?

- JavaScript is a high-level programming language used to make web pages interactive.
- It runs in browsers and also on servers using Node.js.
- Applications include web development, APIs, games, and full-stack applications.

■ Important Question: Variables and Data Types

- Variables store data and are declared using var, let, and const.
- Primitive data types: Number, String, Boolean, Undefined, Null, BigInt, Symbol.
- Non-primitive data types: Objects, Arrays, and Functions.

■ Important Question: Operators and Conditional Statements

- Operators perform operations on values and variables.
- Conditional statements such as if, if-else, and switch control program flow.

■ Important Question: Loops and Strings

- Loops repeat a block of code multiple times.
- JavaScript supports for, while, do-while, for...of, and for...in loops.
- Strings are sequences of characters and provide many useful methods.

■ Important Question: Arrays

- Arrays store multiple values in one variable.
- Important methods include push(), pop(), map(), filter(), and reduce().

■ Important Question: Functions

- Functions are reusable blocks of code.
- JavaScript supports normal functions, arrow functions, and callback functions.

■ Important Question: DOM (Document Object Model)

- The DOM represents an HTML document as a tree of objects.
- JavaScript can add, remove, and modify HTML elements dynamically.

■ Important Question: Events

- Events are actions such as clicks, keyboard input, and form submission.

- Event handling is commonly implemented using `addEventListener()`.

■ Important Question: Classes and Objects

- Objects store data in key-value pairs.
- Classes are blueprints used to create objects and support inheritance.

■ Important Question: Callbacks, Promises, and Async/Await

- Callbacks are functions passed into other functions.
- Promises handle asynchronous operations and have Pending, Fulfilled, and Rejected states.
- Async/await simplifies asynchronous programming.

■ Important Question: Fetch API

- Fetch API is used to send HTTP requests and consume APIs.
- It returns a Promise and is commonly used with JSON data.

Most Important Exam Questions

1. What is JavaScript and what are its applications?
2. Differentiate between var, let, and const.
3. Explain primitive and non-primitive data types.
4. Explain operators and conditional statements.
5. Explain different types of loops in JavaScript.
6. Explain arrays and important array methods.
7. What is DOM and how does DOM manipulation work?
8. Explain event handling in JavaScript.
9. Explain classes, objects, and inheritance.
10. What are callbacks, promises, and async/await?
11. Explain Fetch API with an example.